

Sixth Semester B.E./B.Tech. Degree Examination, June/July 2026
Machine Learning

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C																																																
Q.1	a.	Define Machine Learning. Explain different types of machine learning.	10	L2	CO1																																																
	b.	Consider the following set : $S = \{12, 14, 19, 22, 24, 26, 28, 31, 32\}$. Apply various binning techniques and show the result.	6	L3	CO1																																																
	c.	Differentiate between the four types of attributes used in data analysis. Give suitable examples.	4	L2	CO1																																																
OR																																																					
Q.2	a.	Explain machine learning process model with neat diagram and list any two applications of machine learning.	10	L2	CO1																																																
	b.	Consider the set : $V = \{88, 90, 92, 94\}$. Apply min-max procedure and map the marks to a new range 0 – 1.	6	L3	CO1																																																
	c.	Explain any four data visualization techniques used for univariate data analysis.	4	L2	CO1																																																
Module – 2																																																					
Q.3	a.	Demonstrate Find-S algorithm for finding maximally specific hypothesis on the given dataset in Table 3(a). <table border="1"><caption>Table 3(a)</caption><thead><tr><th>Origin</th><th>Manufacture</th><th>Color</th><th>Year</th><th>Type</th><th>Class</th></tr></thead><tbody><tr><td>Japan</td><td>Honda</td><td>Blue</td><td>1980</td><td>Economy</td><td>Yes</td></tr><tr><td>Japan</td><td>Toyota</td><td>Green</td><td>1970</td><td>Sport</td><td>No</td></tr><tr><td>Japan</td><td>Toyota</td><td>Blue</td><td>1990</td><td>Economy</td><td>Yes</td></tr><tr><td>USA</td><td>Audi</td><td>Red</td><td>1980</td><td>Economy</td><td>No</td></tr><tr><td>Japan</td><td>Honda</td><td>White</td><td>1980</td><td>Economy</td><td>Yes</td></tr><tr><td>Japan</td><td>Toyota</td><td>Green</td><td>1980</td><td>Economy</td><td>Yes</td></tr><tr><td>Japan</td><td>Honda</td><td>Red</td><td>1980</td><td>Economy</td><td>No</td></tr></tbody></table>	Origin	Manufacture	Color	Year	Type	Class	Japan	Honda	Blue	1980	Economy	Yes	Japan	Toyota	Green	1970	Sport	No	Japan	Toyota	Blue	1990	Economy	Yes	USA	Audi	Red	1980	Economy	No	Japan	Honda	White	1980	Economy	Yes	Japan	Toyota	Green	1980	Economy	Yes	Japan	Honda	Red	1980	Economy	No	10	L3	CO2
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	b.	Write steps involved in PCA (Principal Component Analysis) algorithm.	5	L2	CO2																																																
	c.	Explain any five metrics used to evaluate the performance of machine learning model.	5	L2	CO2																																																

1 of 3

OR

Q.4	a.	Apply Candidate Elimination Algorithm (CEA) on the following dataset in Table 4(a) and determine the final version space. Table 4(a)	10	L3	CO2																																				
		<table><tr><th>Sl. No. Example</th><th>Sky</th><th>Air Temp</th><th>Humidity</th><th>Wind</th><th>Water</th><th>Forecast</th><th>Enjoy sport</th></tr><tr><td>1</td><td>Sunny</td><td>Warm</td><td>Normal</td><td>Strong</td><td>Warm</td><td>Same</td><td>Yes</td></tr><tr><td>2</td><td>Sunny</td><td>Warm</td><td>High</td><td>Strong</td><td>Warm</td><td>Same</td><td>Yes</td></tr><tr><td>3</td><td>Rainy</td><td>Cold</td><td>High</td><td>Strong</td><td>Warm</td><td>Change</td><td>No</td></tr><tr><td>4</td><td>Sunny</td><td>Warm</td><td>High</td><td>Strong</td><td>Cool</td><td>Change</td><td>Yes</td></tr></table>				Sl. No. Example	Sky	Air Temp	Humidity	Wind	Water	Forecast	Enjoy sport	1	Sunny	Warm	Normal	Strong	Warm	Same	Yes	2	Sunny	Warm	High	Strong	Warm	Same	Yes	3	Rainy	Cold	High	Strong	Warm	Change	No	4	Sunny	Warm	High
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4	Sunny	Warm	High	Strong	Cool	Change	Yes																																		
	b.	Explain Bivariate Statistics. Describe covariance and correlation with formulas and interpretation.	5	L2	CO2																																				
	c.	Explain Re-sampling methods for model evaluation.	5	L2	CO2																																				

Module – 3

Q.5	a.	Using the K-Nearest Neighbors (KNN) algorithm with $K = 3$ and Euclidean distance, classify the result of student having (CGPA = 6.1, Assessment = 40, Project submitted = 5) based on dataset in Table 5(a). Table 5(a)	10	L3	CO3																																														
	<table><tr><th>Sl. No.</th><th>CGPA</th><th>Assessment</th><th>Project Submitted</th><th>Result</th></tr><tr><td>1</td><td>9.2</td><td>85</td><td>8</td><td>Pass</td></tr><tr><td>2</td><td>8</td><td>80</td><td>7</td><td>Pass</td></tr><tr><td>3</td><td>8.5</td><td>81</td><td>8</td><td>Pass</td></tr><tr><td>4</td><td>6</td><td>45</td><td>5</td><td>Fail</td></tr><tr><td>5</td><td>6.5</td><td>50</td><td>4</td><td>Fail</td></tr><tr><td>6</td><td>8.2</td><td>72</td><td>7</td><td>Pass</td></tr><tr><td>7</td><td>5.8</td><td>38</td><td>5</td><td>Fail</td></tr><tr><td>8</td><td>8.9</td><td>91</td><td>9</td><td>Pass</td></tr></table>					Sl. No.	CGPA	Assessment	Project Submitted	Result	1	9.2	85	8	Pass	2	8	80	7	Pass	3	8.5	81	8	Pass	4	6	45	5	Fail	5	6.5	50	4	Fail	6	8.2	72	7	Pass	7	5.8	38	5	Fail	8	8.9	91	9	Pass	
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b.	List and explain different types of regression methods used in machine learning.	4	L2	CO3																																															
c.	Explain the procedure to construct a decision tree using ID_3 algorithm.	6	L2	CO3																																															

OR

Q.6	a.	Explain nearest centroid classifier and classify test instance (6, 5) using dataset in Table 6(a). Show all steps involved. Table 6(a) <table><tr><th>X</th><th>Y</th><th>Class</th></tr><tr><td>3</td><td>1</td><td>A</td></tr><tr><td>5</td><td>2</td><td>A</td></tr><tr><td>4</td><td>3</td><td>A</td></tr><tr><td>7</td><td>6</td><td>B</td></tr><tr><td>6</td><td>7</td><td>B</td></tr><tr><td>8</td><td>5</td><td>B</td></tr></table>	X	Y	Class	3	1	A	5	2	A	4	3	A	7	6	B	6	7	B	8	5	B	10	L3	CO3
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	b.	Explain concept of logistic regression with its mathematical model and application.	4	L2	CO3
	c.	Explain how the C4.5 algorithm constructs a decision tree for dataset with discrete valued attributes.	6	L2	CO3
Module – 4					
Q.7	a.	Explain Naive Bayes algorithm. Mention its applications and limitation.	10	L3	CO4
	b.	Explain simple model of an artificial neuron and the structure of an Artificial Neural Network (ANN). Also describe any four activation functions.	10	L2	CO4
OR					
Q.8	a.	With a diagram of the Perceptron Model, explain the perceptron learning algorithm.	10	L3	CO4
	b.	Explain different types of Artificial Neural Network (ANNs) and list the major challenges associated with ANN.	10	L2	CO4
Module – 5					
Q.9	a.	Explain different hierarchical clustering algorithms and describe linkage criteria used.	10	L3	CO5
	b.	Explain density based clustering with DBSCAN. Mention its advantages.	10	L2	CO5
OR					
Q.10	a.	Explain Reinforcement Learning (RL), its characteristics and challenges.	10	L2	CO5
	b.	Explain the Q-learning algorithm with its steps and formulas.	10	L2	CO5
